USER GUIDE

Grain ERP

Grain Trading & Warehouse Management System — Module Reference Guide

Desktop application (Tauri + React + Supabase) · Companion Android app · Role-based access
Covers every module: master data, transactions, warehouse operations, and automated financial reports.

1. Overview

Grain ERP is a desktop application (with a companion Android app) for running a grain trading and warehousing business — vendor/customer trading, arhti-style commission handling, godown stock, and fully automated accounting. All data is stored in a shared Supabase (PostgreSQL) database, so desktop and mobile always see the same live data.

How the system is organized

The sidebar is grouped into four sections, plus the Dashboard:

Section	What it contains
Master Data	Reusable records every transaction refers back to: Vendors, Customers, Items, Warehouses, Employees, Bank Accounts, Commission Agents, Shareholders.
Transactions	The day-to-day entries that move stock and money: Purchases, Sales, Payments, Receipts, Installment Plans, Commission Entries.
Warehouse Operations	Physical goods-movement records: Stock on Hand, Gate Pass, Kanda Slips, Bardana Ledger, Chadai Charges, Stock Adjustments.
Financial Reports	Everything derived automatically from the above: Party Ledger, Accounts Payable/Receivable, Expenses, Liabilities, Trial Balance, Balance Sheet, Profit & Loss, Breakeven Analysis.

Core principle: nothing is entered twice

Every financial report on this list is **computed automatically** from the transactions you enter — there is no manual bookkeeping. Record a Purchase, and the system automatically: updates the item's weighted-average cost, increases stock in the chosen warehouse, and posts a payable entry to the vendor's ledger. The same logic applies to Sales (receivable + cost of goods sold), Payments/Receipts (reduce balances), and every warehouse-operations module.

Roles & what they can see

Role	Access
Owner / Admin	Full access to every module, including financials and settings.
Accountant	Ledger, Trial Balance, Balance Sheet, P&L, Bank Accounts, AP/AR, Installments, Expenses, Liabilities. No user management.
Warehouse Staff	Stock entry, Gate Pass, Kanda Slips, Bardana Ledger, Chadai Charges, Stock Adjustments. No financial reports.

Sales / Purchase Staff	Vendor/customer transactions (Purchase/Sale entry), Item master (view), basic customer/vendor balance.
Commission Agent	Views only their own commission ledger and assigned transactions.

2. Dashboard

The landing page after login. Gives an at-a-glance snapshot of the whole business without opening any individual module.

Stat tiles

Ten live numbers: Vendors, Customers, Items, Warehouses, Bank Balance, Total Payable, Total Receivable, Unpaid Liabilities, Expenses (This Month), and Overdue Installments.

Charts

- **Stock by Item** — current on-hand quantity per item, across all warehouses.
- **Purchases vs Sales** — daily totals over the last 30 days, one line per side.
- **Expenses by Category** — where operating costs are going.
- **Top Outstanding Balances** — the largest amounts owed, vendors (payable) and customers (receivable) shown together.

Recent Activity

The latest 10 events across Purchases, Sales, Payments, Receipts, and Expenses, newest first — a quick "what just happened" feed.

3. Master Data

These are the building blocks every transaction points back to. Set these up first.

Vendors master data

The businesses/farmers you buy grain from. Each vendor has contact info, an opening balance, a credit limit/payment terms, and an auto-tracked **payable balance** that grows with every Purchase and shrinks with every Payment.

Used by: Purchases, Payments, Bardana Ledger, Party Ledger, Trial Balance/Balance Sheet.

Customers master data

The businesses you sell grain to. Mirrors Vendors, but tracks a **receivable balance** that grows with Sales and shrinks with Receipts.

Used by: Sales, Receipts, Bardana Ledger, Party Ledger, Trial Balance/Balance Sheet.

Items master data

The commodity catalog (e.g. Wheat, Rice, Maize). Each item stores its unit conversion factor (default: 1 mun = 40 kg, configurable per item) and a **weighted-average cost** that the system recalculates automatically every time you record a Purchase.

Used by: Purchases, Sales, Stock on Hand, Stock Adjustments, Gate Pass, Kanda, and every cost-based report (COGS, Trial Balance, Balance Sheet, Breakeven).

Warehouses master data

Godown locations. Every stock movement references a warehouse — stock is tracked per item *per warehouse*, not just company-wide.

Used by: Purchases, Sales, Stock on Hand, Gate Pass, Stock Adjustments, Expenses (optional link).

Employees master data

Staff records: role, contact, salary terms, joining date. Currently informational (payroll automation is a later add-on); can be referenced from Warehouses as the in-charge person.

Bank Accounts master data

Each bank account tracks an opening balance and a **running balance** that updates automatically whenever an Expense, Payment, or Receipt is recorded against it (bank expenses/payments subtract, receipts add).

Used by: Expenses, Payments, Receipts, Balance Sheet, Trial Balance, Dashboard.

Commission Agents master data

Brokers (arhti) who facilitate trades for a commission. Each agent has a rate — either a percentage of the transaction total, or a fixed amount per mun sold/bought.

Used by: Commission Entries (auto-calculates the amount from the agent's rate).

Shareholders master data

Investors in the business, each with a capital investment amount and a profit-sharing ratio (%). A separate ledger records contributions, withdrawals, and profit-share payouts, giving each shareholder a running balance.

Used by: Trial Balance and Balance Sheet (shareholder capital feeds into total equity).

4. Transactions

The entries that actually move stock and money. Every one of these posts automatically to the relevant ledgers — no manual double-entry required.

Purchases

Record buying stock from a vendor: item, quantity (kg or mun — pick the unit and it auto-converts), rate, warehouse received into, and date.

On save, the system automatically:

- Recalculates the item's **weighted-average cost** across all warehouses.
- Increases **stock on hand** in the chosen warehouse.
- Posts a **payable entry** to the vendor's ledger.
- Assigns an **invoice/bill number** and makes a printable bill available.

Deleting a purchase reverses all of the above automatically (stock, average cost, and the ledger entry are unwound).

Sales

Record selling stock to a customer: item, quantity, rate, warehouse dispatched from, date.

On save: reduces stock, computes **cost of goods sold** at the item's current average cost, posts a **receivable entry** to the customer's ledger, and generates a printable invoice. Deleting a sale reverses stock and the ledger entry.

Payments

Record money paid *to* a vendor (full or partial), by cash or bank. Automatically reduces the vendor's payable balance and, if paid from a bank account, reduces that account's balance too.

Receipts

Record money received *from* a customer. Reduces the customer's receivable balance and increases the chosen bank account's balance if received into the bank.

Installment Plans

Works in both directions — payable (to a vendor) or receivable (from a customer). Enter a total amount, number of installments, start date, and frequency (days); the system auto-generates each installment with its due date. Installments are flagged **Overdue** automatically once past due, and can be marked **Paid** with one click.

Commission Entries

Link a commission to an existing Purchase or Sale. Two types:

- **Self** — commission the business itself earns (acting as broker); amount entered manually.
- **Agent** — commission owed to a linked Commission Agent; the amount is **calculated automatically** from the agent's rate (percentage of total, or fixed per mun) the moment the entry is saved.

5. Warehouse Operations

Physical goods-movement records — these track what's happening on the ground, often independent of (but linkable to) the financial Purchase/Sale entry.

Stock on Hand

Real-time read-only view of quantity per item, per warehouse — shown in both kg and mun. This is the single source of truth for "how much do we have," fed by every module that moves stock (Purchases, Sales, Stock Adjustments).

Gate Pass

Logs a vehicle physically entering or leaving a warehouse: vehicle number, driver, item, quantity, direction (in/out), and optionally linked to the Purchase or Sale it corresponds to. This documents physical movement independent of the financial entry — useful for security/audit at the gate.

Kanda (Weighment) Slips

The official weighbridge record backing a Gate Pass: gross weight, tare (empty vehicle) weight, and **net weight** (calculated automatically as gross minus tare).

Bardana Ledger

Tracks empty sacks/bags supplied to or returned from vendors and customers — a separate **quantity** ledger, not cash. Each vendor/customer has a running bag balance so you always know who owes bags to whom.

Chadai Charges

Loading/unloading labor charges, linked to a specific Purchase or Sale. Choose a basis — per bag, per mun, or a lump sum — and the amount is calculated automatically (rate × quantity, or the flat amount for lump sum).

Stock Adjustments

Records weight gain or loss on stored stock over time — moisture loss, shrinkage, handling damage, or a reweighing correction. Pick a direction (increase/decrease), enter the quantity and a reason; stock on hand updates immediately. The system blocks any adjustment that would push stock negative.

6. Financial Reports

Nothing here is entered manually — every figure is derived automatically from the transactions above.

Party Ledger report

Pick any vendor or customer to see their full debit/credit history and running balance, built automatically from every Purchase, Sale, Payment, and Receipt involving them.

Accounts Payable / Receivable report

Live total and per-party breakdown of what the business owes (payable, to vendors) and is owed (receivable, from customers) — always current, no month-end close needed.

Expenses transaction + report

Operating costs by category, date, and payment source (cash or a specific bank account), optionally tagged to a warehouse. Feeds the Dashboard's "Expenses by Category" chart and every profit calculation.

Liabilities transaction + report

A register for obligations outside normal vendor payables — bank loans, unpaid bills, etc. — with a due date and paid/unpaid status.

Trial Balance automated report

Every account's debit and credit totals in one table — Bank, Stock Valuation, Accounts Receivable/Payable, Liabilities, Shareholder Capital, Sales Revenue, Cost of Goods Sold, Expenses, Chadai Charges, and Commission — always balanced. If debits and credits don't naturally tie out, a "Retained Earnings" line automatically absorbs the difference, exactly as standard bookkeeping practice does.

Balance Sheet automated report

A snapshot as of today, in three sections:

- **Assets** — bank balances, stock valuation (at average cost), accounts receivable.
- **Liabilities** — accounts payable, other unpaid liabilities.
- **Equity** — shareholder capital plus retained earnings (net profit to date).

> If Assets don't exactly equal Liabilities + Equity, the page shows a warning explaining why — almost always because an opening balance (for a vendor, customer, or shareholder) was entered without a matching capital/liability entry from before the system went live. This is flagged rather than hidden, so it's an actionable signal, not a silent error.

Profit & Loss automated report

Pick any date range and get a full statement: Sales Revenue – Cost of Goods Sold = Gross Profit, minus Operating Expenses, Chadai Charges, and Commission = Net Profit.

Breakeven Analysis automated report

For a chosen date range: fixed costs (from Expenses), average sale price and average cost per kg (from Sales), and the resulting contribution margin. From that it calculates the **breakeven volume** (kg that must be sold to cover costs) and **breakeven revenue**.

7. Printable Documents

Invoices / Bills

Every Purchase and Sale gets an auto-numbered bill (P-# for purchases, S-# for sales). Open the printer icon next to any row to view a formatted invoice with party details, item, quantity (kg and mun), rate, and total — ready to print directly from the app.